PROJECT REPORT

CURRENCY CONVERTER

Python 3 Console-Based Application

Project Statement No.	25
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Programming Language	Python 3

Submitted as part of the Python programming project work.

1. Introduction

The Currency Converter is a console-based Python application that converts an amount from one supported currency to another. The application supports INR, USD, EUR and GBP. It uses fixed exchange rates supplied in the project statement and routes every conversion through Indian Rupees (INR).

2. Problem Statement

Develop a Python 3 program that accepts an amount, a source currency and a target currency, then displays the converted amount rounded to two decimal places. Exchange rates must be stored in a dictionary. The program must use functions, conditions and operators, and validate currency codes and amount input.

3. Objectives

The main objectives of this project are:

- Convert values between four supported currencies.
- Store exchange rates using a Python dictionary.
- Organise the conversion logic into reusable functions.
- Validate user input and handle common invalid entries.
- Display conversion results to two decimal places.

4. Tools and Technologies

- Programming language: Python 3
- Development environment: Visual Studio Code (VS Code)
- Version control and submission: GitHub
- Python module used for precise decimal arithmetic: decimal

5. Supported Currencies and Exchange Rates

Currency code	Value in INR
INR	1
USD	83
EUR	90
GBP	105

Note: These are fixed project rates, not live market exchange rates.

6. Working Method and Conversion Formula

Every conversion is performed in two stages so that INR acts as the common base currency:

Stage 1: Convert the source amount into INR.

Amount in INR = Source amount × Source currency rate

Stage 2: Convert the INR amount into the target currency.

Target amount = Amount in INR ÷ Target currency rate

The final target amount is rounded to two decimal places before it is displayed.

7. Program Structure

exchange_rates
A dictionary stores each supported currency code and its value in INR.

convert_to_inr(amount, currency)
Multiplies the entered amount by the selected currency's INR rate and returns the equivalent value in INR.

convert_currency(amount, from_currency, to_currency)
Uses the first function to convert the source amount into INR, divides by the target currency rate, and rounds the result to two decimal places.

main()
Takes input from the user, normalises currency codes, validates the amount and currency codes, calls the conversion function, and displays the result.

8. Python Concepts Used

Concept	Use in the project
Dictionary	Stores the fixed exchange rates for INR, USD, EUR and GBP.
Functions	Separates conversion tasks into reusable blocks.
Conditional statements	Checks whether currencies are supported and whether the amount is valid.
Operators	Multiplication and division perform the currency calculations.
Exception handling	Handles input that cannot be converted into a valid number.
Decimal arithmetic	Supports decimal calculations and rounding to two places.

9. Testing and Results

The program was tested with normal conversions, zero amount, unsupported currency, non-numeric input and negative amount. The expected results are listed below. The corresponding terminal screenshots are stored in the **screenshots/** folder in the GitHub repository.

Test	Input	Expected result
1	166 INR → USD	2.00 USD
2	0 USD → INR	0.00 INR
3(a)	Source currency: JPY	Unsupported currency message
3(b)	Amount: abc	Invalid amount message
3(c)	Amount: -50	Amount cannot be negative message
4(a)	100 USD → EUR	92.22 EUR
4(b)	2 GBP → INR	210.00 INR
4(c)	45 EUR → USD	48.80 USD

Screenshot evidence: The individual test-case screenshots are maintained in the repository's screenshots folder. They can be viewed alongside the source code on GitHub.

10. Conclusion

The Currency Converter project demonstrates how fundamental Python concepts can be combined to build a useful console application. It uses a dictionary to store exchange rates, functions to organise conversion logic, conditions to validate inputs, operators to perform calculations, and exception handling to manage invalid numerical input. The project also provided practical experience in testing a program and maintaining project files using GitHub.

Repository contents

Currency_Converter.py — source code
README.md — project overview and run instructions
screenshots/ — terminal evidence for the test cases